

(22) $\rightarrow$

$$\begin{aligned} & \frac{1}{16} \pi^2 \left(-\frac{1}{18} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{3,0}[m_2] + \frac{1}{12} (-g_1^2 g_2^4 g_2^6) \text{LF}_{4,-1}[m_2] + \frac{4}{45} g_2^4 (g_1^2 - g_2^2) \text{LF}_{5,-2}[m_2] + \right. \\ & \frac{1}{72} \sum_p g_1^6 \text{LF}_{3,0}[m_p^P] - \frac{1}{6} g_1^4 \overline{y_d^{PR}} y_d^{PR} \text{LF}_{3,0}[m_d^r] + \\ & \frac{1}{8} \sum_p g_1^6 \text{LF}_{3,0}[m_p^P] - \frac{1}{2} g_1^4 \overline{y_e^{PR}} y_e^{PR} \text{LF}_{3,0}[m_e^r] - \\ & \frac{1}{288} (g_1^2 - g_2^2) (36 \overline{y_e^{PR}} y_e^{PR} (g_1^2 + 3 g_2^2) + \sum_p (9 g_1^4 + 9 g_1^2 g_2^2 - 10 g_2^4)) \text{LF}_{3,0}[m_1^P] + \\ & \frac{1}{96} g_2^2 (-g_1^2 + g_2^2) (-24 \overline{y_e^{PR}} y_e^{PR} + \sum_p g_2^2) \text{LF}_{4,-1}[m_1^P] + \\ & \frac{1}{45} \sum_p g_2^4 (-g_1^2 + g_2^2) \text{LF}_{5,-2}[m_1^P] + \frac{1}{288} (-12 \overline{y_d^{PR}} y_d^{PR} (g_1^4 + 18 g_1^2 g_2^2 - 27 g_2^4) + \\ & 6 \overline{y_u^{PR}} y_u^{PR} (g_1^2 + 3 g_2^2)^2 + \sum_p (g_1^6 + 21 g_1^2 g_2^4 - 30 g_2^6)) \text{LF}_{3,0}[m_q^P] + \\ & \frac{1}{32} g_2^2 (-g_1^2 + g_2^2) (-24 \overline{y_d^{PR}} y_d^{PR} + \sum_p g_2^2) \text{LF}_{4,-1}[m_q^P] + \frac{1}{15} \sum_p g_2^4 (-g_1^2 + g_2^2) \text{LF}_{5,-2}[m_q^P] - \\ & \frac{1}{9} \sum_p g_1^6 \text{LF}_{3,0}[m_u^P] + \frac{1}{3} g_1^4 \overline{y_u^{PR}} y_u^{PR} \text{LF}_{3,0}[m_u^r] + \frac{1}{36} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{3,0}[\tilde{\mu}] + \\ & \frac{1}{24} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{4,-1}[\tilde{\mu}] + \frac{2}{45} g_2^4 (g_1^2 - g_2^2) \text{LF}_{5,-2}[\tilde{\mu}] - \\ & \frac{1}{4} g_1^6 \text{LF}_{3,-3}[m_1, \tilde{\mu}] - \frac{3}{4} g_1^6 m_1^2 \text{LF}_{3,-3}[m_1, \tilde{\mu}] + \frac{1}{3} g_2^4 (g_1^2 - g_2^2) \text{LF}_{2,1,0}[m_2, \tilde{\mu}] + \\ & \frac{17}{24} g_2^4 (g_1^2 - g_2^2) \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] + \frac{1}{6} (-g_1^2 g_2^4 + g_2^6) \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \\ & \frac{7}{6} g_2^4 (-g_1^2 + g_2^2) \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] + \frac{3}{4} g_2^4 m_2^2 (g_1^2 - g_2^2) \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] + \\ & \frac{1}{4} g_2^4 (2 g_1^2 - 11 g_2^2) \text{LF}_{3,3,-3}[m_2, \tilde{\mu}] - \frac{1}{4} g_2^4 m_2^2 (2 g_1^2 + g_2^2) \text{LF}_{3,3,-2}[m_2, \tilde{\mu}] + \\ & \frac{1}{2} g_2^4 (g_1^2 - g_2^2) \text{LF}_{4,2,-3}[m_2, \tilde{\mu}] + \frac{1}{2} g_2^4 m_2^2 (-g_1^2 + g_2^2) \text{LF}_{4,2,-2}[m_2, \tilde{\mu}] + \\ & \frac{1}{2} g_1^2 \overline{y_d^{PR}} y_d^{PR} y_d^{PR} y_d^{PR} \text{LF}_{2,1,0}[m_d^r, m_d^t] - \frac{1}{12} g_1^4 (4 \overline{a_d^{PR}} a_d^{PR} + \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{2,2,0}[m_d^r, m_q^P] + \\ & \frac{1}{12} g_1^4 (-2 \overline{a_d^{PR}} a_d^{PR} + \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{3,1,0}[m_d^r, m_q^P] - \\ & \frac{1}{2} g_1^2 \overline{y_d^{PR}} y_d^{PR} \overline{y_u^{PR}} y_u^{PR} \text{LF}_{2,1,0}[m_d^r, m_u^t] + \frac{1}{2} g_1^2 \overline{y_e^{PR}} \overline{y_e^{PR}} y_e^{PR} y_e^{PR} \text{LF}_{2,1,0}[m_e^r, m_e^t] - \\ & \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR} \text{LF}_{2,2,0}[m_e^r, m_1^P] + \frac{1}{4} g_1^4 (-2 \overline{a_e^{PR}} a_e^{PR} + \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{3,1,0}[m_e^r, m_1^P] + \\ & \frac{1}{4} (2 g_2^4 \overline{a_e^{PR}} a_e^{PR} + g_1^4 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) \text{LF}_{3,1,0}[m_1^P, m_e^r] + \\ & \frac{1}{8} (2 g_1^4 \overline{a_e^{PR}} a_e^{PR} + g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR} (g_1^2 - g_2^2)) \text{LF}_{3,2,-1}[m_1^P, m_e^r] + \\ & \frac{1}{8} (-2 g_2^2 \overline{a_e^{PR}} a_e^{PR} (g_1^2 + 2 g_2^2) + 3 g_1^2 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR} (-g_1^2 + g_2^2)) \text{LF}_{4,1,-1}[m_1^P, m_e^r] - \\ & \frac{1}{24} (g_1^2 - g_2^2) (2 \overline{a_e^{PR}} a_e^{PR} (3 g_1^2 + g_2^2) + 3 \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR} (-g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[m_1^P, m_e^r] - \\ & \frac{1}{2} g_2^2 \overline{y_e^{PR}} \overline{y_e^{PR}} y_e^{PR} y_e^{PR} \text{LF}_{2,1,0}[m_1^P, m_1^S] + \\ & \frac{1}{4} \overline{y_e^{PR}} \overline{y_e^{PR}} y_e^{PR} y_e^{PR} (-g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_1^P, m_1^S] + \\ & \frac{1}{12} (2 \overline{a_d^{PR}} a_d^{PR} (-4 g_1^4 + 9 g_2^4) + g_1^4 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) \text{LF}_{3,1,0}[m_q^P, m_d^r] + \\ & \frac{1}{8} (2 g_1^4 \overline{a_d^{PR}} a_d^{PR} + g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR} (g_1^2 - g_2^2)) \text{LF}_{3,2,-1}[m_q^P, m_d^r] + \\ & \frac{3}{8} (2 \overline{a_d^{PR}} a_d^{PR} (2 g_1^4 - g_1^2 g_2^2 - 2 g_2^4) + g_1^2 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR} (-g_1^2 + g_2^2)) \text{LF}_{4,1,-1}[m_q^P, m_d^r] - \\ & \frac{1}{8} (g_1^2 - g_2^2) (2 \overline{a_d^{PR}} a_d^{PR} (3 g_1^2 + g_2^2) + 3 \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR} (-g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[m_q^P, m_d^r] + \\ & \frac{1}{4} \overline{y_d^{PR}} y_d^{PR} (2 \overline{y_d^{PR}} y_d^{PR} (2 g_1^2 - 3 g_2^2) - \overline{y_u^{PR}} y_u^{PR} (g_1^2 + 3 g_2^2)) \text{LF}_{2,1,0}[m_q^P, m_q^S] + \\ & \frac{3}{4} \overline{y_d^{PR}} \overline{y_d^{PR}} y_d^{PR} y_d^{PR} (-g_1^2 + g_2^2) \text{LF}_{3,1,-1}[m_q^P, m_q^S] + \\ & \frac{1}{48} (\overline{a_u^{PR}} a_u^{PR} (-7 g_1^4 - 18 g_1^2 g_2^2 + 9 g_2^4) + 2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^4 - 6 g_1^2 g_2^2 - 3 g_2^4)) \\ & \text{LF}_{2,2,0}[m_q^P, m_u^r] + \frac{1}{48} (\overline{a_u^{PR}} a_u^{PR} (g_1^2 + 3 g_2^2)^2 - 2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^4 - 18 g_1^2 g_2^2 + 9 g_2^4)) \\ & \text{LF}_{3,1,0}[m_q^P, m_u^r] + \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (-g_1^2 + g_2^2) \text{LF}_{3,2,-1}[m_q^P, m_u^r] + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} \\ & (-g_1^2 + g_2^2) \text{LF}_{4,1,-1}[m_q^P, m_u^r] - \frac{1}{4} \overline{y_d^{PR}} y_d^{PR} \overline{y_u^{PR}} y_u^{PR} (g_1^2 + 3 g_2^2) \text{LF}_{2,1,0}[m_q^S, m_q^P] + \\ & \frac{1}{48} (\overline{a_u^{PR}} a_u^{PR} (7 g_1^2 - 3 g_2^2)^2 - 2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^4 - 6 g_1^2 g_2^2 - 3 g_2^4)) \text{LF}_{3,1,0}[m_u^r, m_q^P] + \\ & \frac{1}{16} (\overline{a_u^{PR}} a_u^{PR} (g_1^4 + 2 g_1^2 g_2^2 - 3 g_2^4) + 2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^2 + g_2^2)^2) \text{LF}_{3,2,-1}[m_u^r, m_q^P] - \\ & \frac{3}{16} (\overline{a_u^{PR}} a_u^{PR} (7 g_1^4 - 10 g_1^2 g_2^2 + 3 g_2^4) + 2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (g_1^2 + g_2^2)^2) \text{LF}_{4,1,-1}[m_u^r, m_q^P] + \\ & \frac{1}{8} (g_1^2 - g_2^2) (3 \overline{a_u^{PR}} a_u^{PR} (g_1^2 - g_2^2) - 2 \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR} (3 g_1^2 + g_2^2)) \text{LF}_{5,1,-2}[$$